SIA SOFTWARE DEVELOPMENT · GROUP 4 DELIVERABLE

STORA: Draw It, Then Prove It

Comprehensive Speaker Notes, Technical Defense Guide & Full Terminology Master Reference for SIA Technical Panel Defense.

PROJECT	TOPICS COVERED	DECK SIZE	LIVE ARTEFACTS
STORA Retail OS	13, 14, 15, 16	19 Widescreen Slides	6 ADRs, UAT, A11Y Audit

📖 Master Reference: Technical Terms, Full Words & Their Purpose

Use these full names and clear purposes when speaking to the panel:

TERM	FULL WORDS / EXPANSION	CONCRETE PURPOSE IN STORA
`UML`	**Unified Modeling Language**	Standardized visual blueprint language used to model software use-cases, sequence flows, and database schemas so all developers share the same architectural model.
`C4 Model`	**Context, Containers, Components, Code**	A 4-tier zoom-level framework (by Simon Brown) that visualizes software systems like Google Maps: from high-level actors (Context) to deployable units (Containers).
`ADR`	**Architecture Decision Record**	A 1-page immutable document capturing a major technical choice, its constraints, and trade-offs, preserving why a framework or database was chosen.
`UAT`	**User Acceptance Testing**	The final human-testing phase where 5 real sari-sari store owners and 5 neighborhood shoppers complete tasks to prove the app is usable in the real world.
`RA 10173`	**Republic Act No. 10173 (Data Privacy Act of 2012)**	The Philippine national privacy law mandating written informed consent, data minimization, and user right to erasure during testing and live operations.
`a11y`	**Accessibility** (11 letters between 'a' and 'y')	Designing software usable by everyone—including elderly sari-sari store owners, users with low vision, motor difficulties, or those operating under tropical sunlight.
`WCAG`	**Web Content Accessibility Guidelines**	The international W3C accessibility benchmark establishing testable criteria (e.g. 4.5:1 minimum contrast, 44×44dp touch target dimensions).
`ERD`	**Entity-Relationship Diagram**	A visual data model defining database entities (tables), fields, and foreign keys, ensuring data integrity before writing database migrations.
`ACID`	**Atomicity, Consistency, Isolation, Durability**	Four database guarantees in PostgreSQL ensuring community store credit (*utang*) and payments cannot suffer from phantom balances or corrupted records.
`POS`	**Point of Sale**	The retail checkout terminal interface where store owners scan items via camera, calculate change (*sukli*), record payments, and deduct inventory.
`REST API`	**Representational State Transfer App Programming Interface**	A standardized, stateless HTTP communication protocol allowing Flutter mobile apps to query and send data to the Django server using JSON payloads.

TERM	FULL WORDS / EXPANSION	CONCRETE PURPOSE IN STORA
JWT	**JSON Web Token**	A cryptographically signed token enabling stateless user authentication between the mobile phone and backend server without database lookups on every request.

SLIDE 01 TITLE & HIGH-LEVEL FRAMING Foundation

Draw it, then prove it

🎯 SLIDE PURPOSE & CORE MEANING

Establishes the central thesis: **Running code is not enough**. A complete software system must be clearly diagrammed so others can understand it, technical choices must be justified with written evidence, and usability must be proven with real sari-sari store owners and shoppers under Philippine law.

📖 TECHNICAL TERMS DECODED FOR SLIDE 1

- **UML (Unified Modeling Language):** The industry-standard visual modeling notation used to create blueprints of software architecture and database models so teams don't rely on confusing spoken descriptions.
- **C4 Model (Context, Containers, Components, Code):** An architectural diagramming framework that zooms like Google Maps from the high-level business ecosystem down to code classes.
- **ADR (Architecture Decision Record):** A standardized, 1-page document capturing an important architectural decision and its trade-offs, frozen in time so future maintainers understand why the choice was made.
- **UAT (User Acceptance Testing):** The critical pre-release phase where real users (not the programmers) test the software to verify that it meets practical business needs.
- **RA 10173 (Republic Act No. 10173 - Data Privacy Act of 2012):** The Philippine national law governing personal data handling, requiring informed consent and data protection before user testing.
- **POS (Point of Sale):** The digital cash register system where barcode scanning, sales calculation, and stock deduction occur.

🎙️ PRESENTER SCRIPT (VERBATIM WHAT TO SAY)

SPOKEN GUIDE:

"Good morning / afternoon members of the panel. Today, Group 4 presents 'Draw it, then prove it' for STORA—our smart inventory and Point of Sale (POS) retail operating system. In software engineering, writing code that runs on your local laptop is only the first step. True engineering maturity requires proving three critical things: First, can we explain our architecture in under 30 seconds using Unified Modeling Language (UML) and the C4 Model? Second, can we justify every major design choice through Architecture Decision Records (ADRs) rather than making up answers on the spot? And third, can real community sari-sari store owners and shoppers actually use it under Philippine Data Privacy law (RA 10173)? Over this presentation, we will walk you through the architectural blueprints, decision records, empirical user testing, and accessibility fixes that prove STORA is ready for production."

⚠️ POTENTIAL PANEL QUESTION & BULLETPROOF ANSWER

QUESTION: "WHY IS GROUP 4 FOCUSED ON 'DRAWING' AND 'PROVING' INSTEAD OF ADDING MORE FEATURES TO THE APP?"

Answer: "Because adding features to an undocumented, unverified system creates dangerous technical debt. If a developer leaves the project, undocumented decisions lead to expensive rewrites. Group 4 ensures STORA is understandable to any new engineer, legally compliant under RA 10173, and empirically proven with real human beings before deployment."

💡 DELIVERABLES CITED:

Topic 13 (C4/UML Diagrams), Topic 14 (6 ADRs in `docs/adr/`), Topic 15 (UAT Plan & RA 10173 Consent Form in `docs/uat/`), and Topic 16 (WCAG 2.1 AA Accessibility Audit in `docs/accessibility/`).

SLIDE 02 PIPELINE ROLE & MANDATE Team Ownership

Four topics: make the system legible, then validated

🎯 SLIDE PURPOSE & CORE MEANING

Defines where Group 4 sits in the 5-group software engineering pipeline. Groups 1–3 built the backend, database, and client apps. Group 4's job is not rewriting their code, but acting as the critical quality, documentation, and human-validation bridge before Group 5 deploys it.

📖 TECHNICAL TERMS DECODED FOR SLIDE 2

- **C4 & UML Diagrams (Topic 13):** Creating visual system representations at the correct abstraction level so teachers, evaluators, and engineers can audit the system without reading thousands of lines of code.
- **Architecture Decision Records / ADRs (Topic 14):** Capturing the technical rationale for choices like Django REST Framework over GraphQL, PostgreSQL over MongoDB, and dual Flutter clients.
- **User Acceptance Testing / UAT & Consent (Topic 15):** Testing with real merchants and consumers under signed Republic Act No. 10173 informed consent forms.
- **Accessibility / a11y (Topic 16):** Ensuring the app adheres to Web Content Accessibility Guidelines (WCAG) so it functions in direct tropical sunlight, for low-vision users, and with TalkBack screen readers.
- **API (Application Programming Interface):** The structured communication gateway between mobile frontends and the Django backend.

🎙️ PRESENTER SCRIPT (VERBATIM WHAT TO SAY)

SPOKEN GUIDE:

"To understand Group 4's responsibility, look at our development pipeline: Groups 1, 2, and 3 constructed the technical foundation—the Application Programming Interfaces (APIs), the database tables, and the Flutter screens. Group 4 owns legibility and human validation across four mandatory topics: Topic 13 gives us the visual blueprints through Unified Modeling Language (UML) and C4 diagrams. Topic 14 freezes our architectural reasoning into Architecture Decision Records (ADRs). Topic 15 tests the app with real human beings under Republic Act No. 10173 data privacy consent. Topic 16 guarantees accessibility compliance across lighting conditions and physical abilities. Group 5 cannot deploy STORA to Google Play until Group 4 signs off that our criteria are satisfied."

⚠️ POTENTIAL PANEL QUESTION & BULLETPROOF ANSWER

QUESTION: "WHAT HAPPENS IF GROUP 4 FINDS MAJOR USABILITY FLAWS DURING UAT?"

Answer: "That is precisely why Group 4 exists as a gate before Group 5. If our User Acceptance Testing uncovers friction—for example, if merchants struggle to find the cash payment button—we log the issue, work with the development team to refine the interface, and re-test before Group 5 executes production deployment."

💡 TEAM PIPELINE HANDOFF:

Groups 1–3 (Build & Wire) ➔ **Group 4 (Legibility & Human Validation)** ➔ Group 5 (Production Deployment & Security).

SLIDE 03 STRATEGIC VALUE Engineering Philosophy

A system nobody can explain is one nobody can trust

🎯 SLIDE PURPOSE & CORE MEANING

Breaks down the three essential verification verbs: **SHOW IT** (pictures beat 1,000 words), **JUSTIFY IT** ("Why did you...?" has a documented answer), and **PROVE IT** (real users with disabilities and low tech-literacy, not internal team members).

📖 TECHNICAL TERMS DECODED FOR SLIDE 3

- **C4 Container Architecture (Show It):** Replacing confusing verbal descriptions with a clean diagram showing how `stora_owner`, `stora_customer`, Django REST API, and PostgreSQL on Render communicate over HTTPS.
- **ADR / Architectural Trade-offs (Justify It):** Providing written evidence for technical choices (REST vs GraphQL, PostgreSQL vs NoSQL) so evaluators see that choices were intentional, not accidental.
- **UAT Empirical Evidence (Prove It):** Replacing subjective claims (*"we think our app is intuitive"*) with hard empirical data (*"5 of 5 merchants finished checkout in 54 seconds unaided"*).
- **Motor & Visual Disabilities (Accessibility):** Ensuring touch targets meet the 44×44dp standard (STORA achieved 54dp) and text meets 4.5:1 contrast (STORA achieved 15.8:1).

🎙️ PRESENTER SCRIPT (VERBATIM WHAT TO SAY)

SPOKEN GUIDE:

"A working system that nobody can explain is a black box that nobody can trust. When an evaluator or investor asks, 'How does your checkout handle network failure?', talking for ten minutes in prose confuses the room. A single C4 container diagram answers it in 30 seconds. When the panel asks, 'Why didn't you use MongoDB or GraphQL?', pulling up ADR-0001 and ADR-0002 proves our choices were deliberate engineering trade-offs made months ago. And when someone asks if the app is easy to use, our personal opinion doesn't matter. Five real sari-sari store owners successfully completing a sale in 54 seconds is objective proof."

⚠️ POTENTIAL PANEL QUESTION & BULLETPROOF ANSWER

QUESTION: "CAN'T A GOOD DEVELOPER JUST EXPLAIN THEIR CODE DURING THE ORAL DEFENSE WITHOUT DIAGRAMS OR ADRS?"

Answer: "Spoken explanations fade, but written Architecture Decision Records and standardized C4 diagrams persist. In industry, software is maintained by engineers who were never in the original room. Writing down the 'why' and diagramming the 'what' is the hallmark of a professional software engineer."

💡 DELIVERABLES CITED:

Mermaid Container Diagram in `ARCHITECTURE.md`, ADR-0001 & ADR-0002 in `docs/adr/`, and UAT Results Table in `docs/uat/UAT_PLAN_AND_CRITERIA.md`.

SLIDE 04 & 05 TOPIC 13: C4 ARCHITECTURAL MODEL Architecture

Four zoom levels — show only what the audience needs

🎯 SLIDE PURPOSE & CORE MEANING

Explains the four zoom levels of Simon Brown's C4 model (Context, Container, Component, Code) and how STORA maps to each:

- **1. System Context:** STORA as a single box surrounded by Store Owners, Shoppers, GCash QR, and Firebase Cloud Messaging (FCM). For non-technical stakeholders.
- **2. Container:** High-level deployable units: `stora_owner` (Flutter), `stora_customer` (Flutter), Django REST API, and PostgreSQL on Render.
- **3. Component:** Inside the backend: Authentication handler, Inventory serializer, POS Checkout service, and FCM dispatchers.
- **4. Code:** Individual class hierarchies and methods (auto-generated by IDE, omitted from manual hand-drawing).

📖 **TECHNICAL TERMS DECODED:**

C4 Model: Context, Containers, Components, Code. **FCM:** Firebase Cloud Messaging (push notifications). **REST:** Representational State Transfer.

SPOKEN GUIDE:

"Think of C4 like Google Maps: Level 1 (Context) is the continent view—who uses STORA and what external systems like GCash and Firebase touch it? Level 2 (Container) is the city view—what are our deployable containers? We have two Flutter mobile clients communicating over HTTPS with a Django REST backend and a managed PostgreSQL database. For our defense, Levels 1 and 2 are all the panel needs to understand our entire operational landscape."

SLIDE 06 TOPIC 13: UML SELECTION UML Modeling

Three UML diagrams earn their keep — pick by the question

🎯 SLIDE PURPOSE & CORE MEANING

Rejects the antipattern of drawing all 14 Unified Modeling Language (UML) diagrams just to fill pages. Focuses on the three that actually provide value:

- **Use-Case Diagram ("Who can do what?"):** Maps store owner actions (barcode scanning, stock adjustments, utang tracking) vs customer actions (order placement, store radar).
- **Sequence Diagram ("What talks to what, in what order?"):** Essential for tracing tricky financial flows like Fast Checkout: Client ➔ Local Cache ➔ Django ➔ Postgres Stock Lock ➔ FCM.
- **Class / Entity-Relationship Diagram (ERD) ("How is data structured?"):** Maps relational tables: User, StoreLocation, Product, Category, Order, and UtangLedger.

SPOKEN GUIDE:

"Instead of drawing every UML diagram, we focus on three that earn their keep: Use-Case sets our user scope; Sequence traces our multi-step checkout and stock deduction; and our Entity-Relationship Diagram (ERD) reflects the real PostgreSQL schema created by Group 3. Most importantly: our diagrams match our actual working code 1-to-1."

SLIDE 07 & 08 TOPIC 14: ADR STANDARD ADRs

One short page per decision — four headings, that is all

🎯 SLIDE PURPOSE & CORE MEANING

Deconstructs the standard Michael Nygard Architecture Decision Record format into four mandatory sections:

- **1. Status:** Proposed, Accepted, or Superseded.
- **2. Context:** The technical problem and environmental constraints (e.g. Philippine 3G/4G connectivity, offline store realities).
- **3. Decision:** The explicit, unambiguous choice made (e.g. Django REST Framework).
- **4. Consequences:** Both benefits gained and trade-offs/costs accepted.

> **SPOKEN GUIDE:**
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> *"An ADR is not a 50-page dissertation. It is exactly one page with four headings: Status, Context, Decision, and Consequences. Look at ADR-0001: We chose Django REST Framework over GraphQL. Why? Because sari-sari stores operate on throttled prepaid cellular data. REST enables standard HTTP caching headers and is faster to inspect with standard tooling. If our requirements change later, we don't delete this ADR—we write a new one that supersedes it."*

SLIDE 09 TOPIC 14: ADRS IN PRACTICE Production ADRs

Record the decisions that were hard to make or hard to reverse

🎯 SLIDE PURPOSE & CORE MEANING

Provides the acid test for what warrants an ADR vs what is trivial clutter. Lists STORA's 6 production ADRs:

- **ADR-0001:** REST over GraphQL (predictable caching & network efficiency).
- **ADR-0002:** PostgreSQL over NoSQL (strict ACID compliance for community debt / utang ledgers).
- **ADR-0003:** Dual Flutter Apps (independent Owner and Customer APKs for fast install times and clean role isolation).
- **ADR-0004:** Drift/SQLite Local Caching (guarantees offline POS checkout resilience during power outages).
- **ADR-0005:** RA 10173 Compliance (multi-tenant role isolation and permanent account deletion cascading).
- **ADR-0006:** Hybrid JWT + Firebase Auth (decoupled session tokens with cloud push notifications).

> **SPOKEN GUIDE:**
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> *"What deserves an ADR? Anything that would take weeks to change if we were wrong. For instance, choosing PostgreSQL over MongoDB is an ADR because sari-sari store credit—utang—demands strict ACID transactional consistency; you cannot have phantom balances or partial debits. Conversely, variable names or button colors do not get ADRs. We have 6 official ADRs documented in docs/adr/ covering our entire architectural core."*

SLIDE 10 & 11 TOPIC 15: UAT PROTOCOL Human Testing

Give real users a real task, then be quiet and watch

🎯 **SLIDE PURPOSE & CORE MEANING**

Details the **Silent Observation Protocol** and pre-agreed acceptance criteria:

- **Target Cohort:** 5 real sari-sari store owners and 5 neighborhood shoppers (never test with the developers who wrote the code).
- **Realistic Tasks:** "Scan 2 products and ring up a cash payment with ₱500 bill" or "Find a neighborhood store and order pan de sal."
- **Rule of Silence:** Observers must never coach or say "tap that button". The user's hesitation is critical usability data.
- **Pre-Agreed Acceptance Criteria:** Target set beforehand: *"4 of 5 merchants complete fast checkout unaided in under 90 seconds."*
- **Actual STORA Result: 5 of 5 merchants passed, averaging 54 seconds**.

> **SPOKEN GUIDE:**
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> *"In our UAT, we followed the strict Silent Observation Protocol. We handed the phone to real sari-sari owners, gave them a single goal, and stayed completely silent. We did not rescue them. If they hesitated at checkout, that was our bug to fix. Crucially, we set our pass criteria in advance: 4 out of 5 merchants in under 90 seconds. All 5 completed it unaided in an average of 54 seconds, turning our claim into an objective, verifiable fact."*

SLIDE 12 TOPIC 15: ETHICS & LAW RA 10173

Testing with people is a privilege — ask first

🎯 **SLIDE PURPOSE & CORE MEANING**

Applies **Republic Act No. 10173 (Philippine Data Privacy Act of 2012)** to the software testing process. Establishes the Four Pillars of Informed Consent:

- **1. Task Description:** Explicitly state what they will do and the expected duration (10–15 mins).
- **2. Data Collected:** Strictly limited to usability timings and friction logs; zero credit card numbers or financial credentials collected.
- **3. Voluntary Participation:** Participants can stop at any time without penalty or explanation.
- **4. Confidentiality & Right to Erasure:** All participant data is anonymized/de-identified and can be permanently deleted upon request.

> **SPOKEN GUIDE:**
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> *"Testing with real people is not an entitlement—it is a privilege. Under Republic Act 10173, the Philippine Data Privacy Act, you cannot collect user interaction data without prior informed consent. We created a standardized, 1-page consent form in docs/uat/INFORMED_CONSENT_FORM_RA10173.md. Every single participant read and signed this form before touching the app. This demonstrates professional engineering ethics and legal compliance."*

SLIDE 13 & 14 TOPIC 16: ACCESSIBILITY STANDARDS Accessibility

A few habits cover most of your users

🎯 **SLIDE PURPOSE & CORE MEANING**

Presents the 5 high-value fixes implemented in STORA based on Web Content Accessibility Guidelines (WCAG 2.1 AA) and their real human impact:

- **1. Contrast (4.5:1 WCAG AA baseline):** STORA achieved **15.8:1 text contrast** so screens are readable under direct noon tropical sun.
- **2. Touch Targets (44×44 dp minimum):** Re-engineered settings and buttons (`_MenuTile` is **54dp tall**) for shaky hands, older adults, or users riding bumpy jeepneys.
- **3. Semantic Labels:** Every `IconButton` has explicit `tooltip` and `semanticLabel` for Android TalkBack screen readers.
- **4. Not Color Alone:** Form errors explicitly print the word "Required" rather than just turning the border red (supports the ~8% of males with color vision deficiency).
- **5. Scalable Text:** Flexible Flutter layouts adapt dynamically when system font scaling is increased up to 200%.

> **SPOKEN GUIDE:**
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> *"You cannot solve every edge case in one sprint, but these five core habits remove 90% of user barriers. We increased our contrast ratio to 15.8:1, expanded our touch targets to 54dp, added semantic labels to all icon buttons, ensured errors don't rely on red color alone, and validated that layouts survive large text scaling. Accessibility is like curb cuts on sidewalks: built for wheelchairs, but used by everyone with strollers and luggage."*

SLIDE 15 TOPIC 16: TESTING & HONEST GAP Honesty

You can check the basics in fifteen minutes

🎯 **SLIDE PURPOSE & CORE MEANING**

Shows how easily accessibility can be audited on standard mobile devices without expensive enterprise software. Introduces **The Honest Engineering Gap**.

> **SPOKEN GUIDE:**
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> *"Testing accessibility does not require expensive labs. We turned on Android TalkBack, closed our eyes, and checked if we could navigate STORA purely through voice cues. We verified our 15.8:1 contrast using standard tools. And most importantly, we bring evidence rather than false claims: We fixed our settings touch targets and contrast, and we openly state our remaining gap: 'Adding audio feedback beeps during camera barcode scanning is our next planned release.' Evaluators value teams who acknowledge their roadmap honestly."*

SLIDE 16 & 17 DEFENSE EXECUTION PLAYBOOK Playbook

The Demo: Explain it, justify it, and prove real people can use it

🎯 **SLIDE PURPOSE & CORE MEANING**

The Four-Move Demo Playbook for defense day:

- **Move 1 (Diagram):** In 30 seconds, orient the panel with the C4 Container Diagram (Owner App, Customer App, Django, PostgreSQL).
- **Move 2 (ADR):** Read one critical ADR aloud (e.g. ADR-0002: PostgreSQL for utang ACID compliance).
- **Move 3 (UAT):** Present the quantitative result: 5 of 5 store owners in 54s vs 90s criteria, showing the signed RA 10173 form.
- **Move 4 (Accessibility):** Demonstrate 15.8:1 contrast, 54dp targets, TalkBack narration, and state our next milestone.

> **SPOKEN GUIDE:**
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> *"During defense, we don't wander randomly through screens. We execute four disciplined moves: 30 seconds on the container diagram, reading one ADR aloud, presenting our measured UAT numbers with signed consent, and demonstrating our accessibility fixes. This turns a subjective question into an evidence-based defense."*

SLIDE 18 & 19 SCORECARD & HAND-OFF Conclusion

A working system, now explained, justified, and proven

🎯 **SLIDE PURPOSE & CORE MEANING**

Presents the 20-point rubric across all four topics (5 points per topic) that evaluators grade against. Formally hands off the deployment pipeline to Group 5 (Day 23: Security, Scalability, Deployment, and Rollback plans).

> **SPOKEN GUIDE:**
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> *"To conclude: STORA is no longer just working code—it is an explained, justified, and proven platform. All 20 criteria across Diagrams, ADRs, UAT, and Accessibility are satisfied. We hand over our 6 Architecture Decision Records, C4 blueprints, User Acceptance Testing findings, consent records, and accessibility audit to Group 5. They now own the deployment button to ship STORA safely to the public. Thank you, and we welcome your questions."*